

Minwoo Yoo

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Education

The George Washington University

Sep 2022 – May 2027 (expected)

Ph.D. in Economics

Research Areas: Political Economy, Applied Microeconomics, Text Analysis

Seoul National University

M.A. in Economics (Empirical IO) — degree requirements completed

Aug 2020

B.A. in Economics, *summa cum laude* (GPA: 3.93/4.0, Top 5%)

Feb 2018

Working Papers

Trade Shocks and Media Slant [Draft]

- Designed a causal inference strategy exploiting regional differences in trade-shock exposure to estimate the effect of economic shocks on local newspapers' political slant.
- Developed article-level measures of political slant by training supervised text classifiers on U.S. Congressional speeches and applying them to 52M newspaper articles.
- Quantified that the share of Republican-leaning articles rose 9.6 percentage points more in most-exposed regions, driven by initially Democratic-leaning papers in low-education areas.

In and Out: How Do Food Delivery Riders Work? (Draft available upon request)

- Studied how food delivery riders (Uber-like gig workers) adjust participation and hours worked in response to expected wage changes.
- Designed a shift-share IV combining local restaurant composition with national order trends, applied to 21M high-frequency, transaction-level observations.
- Estimated that a 1% wage increase raises hours worked by 2.5–2.8% and the probability of working by 0.7–0.9 percentage points; both responses grow stronger with experience.

Professional Experience

Economic Consultant, Seoul National University

Sep 2018 – Aug 2019

- Prepared analysis submitted to the Korean Fair Trade Commission for the antitrust review of a merger between the world's two largest shipbuilders (Hyundai-Daewoo), with Kim & Chang and Compass Lexecon.
- Built post-merger simulation models of firm-level production and costs; quantified efficiency gains and concluded no expected price increases in Korean government defense contracts.

Projects

Home Credit Default Risk: Imbalanced Classification [GitHub]

Spring 2026

- Built an end-to-end loan default classification pipeline on the Home Credit dataset, covering data preprocessing, feature engineering, model training, and evaluation.
- Compared classical ML models (logistic regression, random forest, gradient boosting) with shallow and deep neural networks, and benchmarked class-imbalance handling techniques such as SMOTE, class weighting, and threshold tuning.

Political Text Classification [GitHub] [Streamlit Demo]

Fall 2025

- Led an end-to-end machine learning project on political text, covering data preprocessing, feature construction, and model training.
- Built and evaluated two separate models: a TF-IDF + Logistic Regression baseline and a fine-tuned transformer-based model, with hyperparameter tuning and performance comparison.

Teaching Experience

The George Washington University

Sep 2023 – Present

Instructor

- Math Camp (Ph.D.), Summers 2024, 2025

Preparatory course for incoming Economics Ph.D. students, covering real analysis, linear algebra, optimization, and statistics. Lectured, designed the syllabus, and wrote the lecture notes, problem sets, and final exam.

Teaching Assistant

- Econometrics I–II (Ph.D.), Fall 2024 – Present
TA for Profs. [Robert F. Phillips](#) and [Ben Williams](#).
- Microeconomic Theory I–II (Ph.D.), Fall 2023 – Spring 2024
TA for Prof. [Sumit Joshi](#).

Seoul National University

Spring 2018 – Fall 2020

Teaching Assistant

- Studies in Industrial Organization (Ph.D.)
- Industrial Organization
- Mathematics for Economics
- Stocks, Bonds, and Financial Derivatives

Skills

Programming & Computing Tools

- Python (pandas, scikit-learn, PyTorch), SQL, R, Stata; AWS, GCP, HPC.

Causal Inference

- Experimental Design (A/B Testing), Difference-in-Differences (DiD), Synthetic Control, Regression Discontinuity Design (RDD), Instrumental Variables (IV), Propensity Score Matching.

Structural Econometrics

- Dynamic Programming, Discrete Choice Models.

Machine Learning

- Logistic Regression, Regularization (Lasso, Ridge), Random Forest, Gradient Boosting, Neural Networks.

Natural Language Processing

- Transformer-based Models (BERT), Seq2Seq with Attention, RNN (LSTM).

References

Leah Brooks (Primary Advisor)

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The George Washington University
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Alessandra Fenizia

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The George Washington University
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